

FPT SCHOOL OF BUSINESS
& TECHNOLOGY

MASTER THESIS ON SOFTWARE ENGINEERING

HYBRID RETRIEVAL AND LEGAL-DOMAIN ADAPTATION FOR VIETNAMESE LEGAL INFORMATION RETRIEVAL

Master's student: NGHIEM PHAN THANH TUAN
Student ID: MSE13189
Supervisor: Dr. Doan Nhat Quang

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Acknowledgements and Declaration of Authorship

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Declaration of Authorship

I declare that this thesis is my own work and has been prepared in accordance with accepted standards of academic integrity. All sources and materials used in this research have been appropriately acknowledged and cited.

The work presented in this thesis has not been submitted, either in whole or in part, for any other degree or academic qualification at this or any other institution.

I take full responsibility for the content, accuracy, and academic integrity of this thesis.

Nghiem Phan Thanh Tuan – 2026

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List of Abbreviations

Terminology follows the retrieval literature and the task definitions introduced in Chapters 3–4 [1, 2, 3, 4].

Abbreviation	Meaning	Explanation
ALQAC	Automated Legal Question Answering Competition	A benchmark and competition for evaluating legal question answering and information retrieval systems.
BCA / MPS	Bộ Công an / Ministry of Public Security collection	A legal question–document collection derived from public questions and published answers associated with Vietnam's Ministry of Public Security.
BM25	Best Matching 25 lexical ranking function	A term-based retrieval method that ranks documents according to query-term matches, term frequency, and document length.
BGE-M3	BGE-M3 multilingual embedding model	A multilingual embedding model used in this thesis as a dense retrieval component.
CV	Cross-validation	An evaluation procedure that partitions data into subsets for model selection and performance estimation.
E5	E5 text-embedding model family	A family of dense embedding models; this thesis evaluates the `multilingual-e5-base` variant.
FT / PT	Fine-tuned / Pretrained	Distinguishes a task-adapted model from its original pretrained version.
IR	Information Retrieval	The task of retrieving and ranking documents that are relevant to a user's information need.
MAP	Mean Average Precision	A ranking metric that averages precision across relevant results and then across queries.
MRR	Mean Reciprocal Rank	A ranking metric based on the reciprocal rank of the first relevant result for each query.
NDCG	Normalized Discounted Cumulative Gain	A ranking metric that rewards relevant documents appearing higher in the ranked result list.

Continued on next page

Abbreviation	**Meaning**	**Explanation**
OOF	Out-of-fold	Predictions or results produced for samples excluded from the corresponding fitting fold.
RAG	Retrieval-Augmented Generation	An architecture that retrieves external information before or during text generation.
RQ	Research Question	A specific question that the thesis investigates empirically.
SD	Standard Deviation	A statistical measure describing dispersion around the mean.

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Chapter 1

Introduction

1.1 Research Context

Retrieval-Augmented Generation (RAG) uses information retrieved from a document collection to support answer generation by a language model [4]. In legal question answering, the retrieved provisions must be relevant to the question. A fluent answer may still be incorrect if those provisions do not support it. This thesis focuses on retrieval effectiveness. Given a Vietnamese legal question, and a corpus containing legal documents that support the anwser, the task is to rank legal articles or documents so that relevant provisions appear near the top of the results.

Vietnamese legal retrieval presents several challenges. Questions often use everyday language, whereas legal provisions use formal terminology. Articles addressing similar subjects may specify different conditions, obligations, or procedures, so similarity in wording or topic alone does not establish relevance. Corpus completeness also affects retrieval: a ranking model cannot retrieve a provision that is absent from the indexed collection. Incorrect mappings between reference provisions and document identifiers can also affect evaluation results. Retrieval evaluation must therefore consider ranking performance, corpus coverage, and the accuracy of these mappings.

BM25 ranks documents using lexical matches and is relevant to legal retrieval because questions may contain terms, document numbers, or phrases found in the corresponding provisions [5, 6]. This thesis uses BM25's evaluation scores as the baseline. Dense retrieval uses embedding similarity to match questions and documents even when their wording differs. The evaluated embedding models are pretrained BGE-M3 and multilingual-E5-base (E5) [2, 3]. The hybrid configurations, BM25+BGE-M3 and BM25+BGE-M3+E5, combine lexical and dense retrieval scores.

The study evaluates these methods on fixed Vietnamese legal datasets using the available relevance annotations. It compares BM25+BGE-M3 with BM25 and tests whether adding E5 improves BM25+BGE-M3. It also examines whether fine-tuning BGE-M3 improves standalone retrieval and whether any gains remain when the model is integrated into a hybrid. Therefore, this thesis contributes to the understanding of hybrid retrieval strategies in the context of Vietnamese legal information retrieval.

1.2 Research Gap and Approach

Prior research provides a basis for the methods evaluated in this thesis. Rosa et al. report that BM25 is a competitive baseline for legal case retrieval [6]. Chen et al. and Wang et al. present BGE-M3 and multilingual E5 and evaluate their embedding models on multilingual benchmarks [2, 3]. Bruch et al. analyze fusion of lexical and semantic retrieval scores, including the effects of normalization and parameter selection [14]. These findings motivate comparisons between lexical retrieval, dense retrieval, and their combinations.

The scope of that evidence also matters. Legal case retrieval differs from retrieving statutory articles for Vietnamese questions. In heterogeneous retrieval evaluation, BEIR identifies limitations in the generalization of dense retrievers across datasets [17]. Zhan et al. show that training with hard negatives can improve dense retrieval on their evaluated benchmarks [16]; their findings concern dense-retriever training rather than the effect of fine-tuning BGE-M3 within the hybrids evaluated here. Therefore, the literature provides a rationale for testing these methods, while their effectiveness on the selected Vietnamese legal collections remains an empirical question.

The research gap addressed here is the evidence needed for two specific decisions: whether adding E5 improves BM25+BGE-M3, and whether fine-tuning BGE-M3 improves both standalone and hybrid retrieval. The thesis tests these questions under defined corpus conditions and separates model selection from evaluation on a separate test set. This is an empirical contribution rather than a new retrieval algorithm. Chapter 2 reviews the supporting literature, and Chapter 4 specifies the comparisons and evaluation procedures.

The approach has three parts. First, ALQAC and Zalo are evaluated using clean reference text and article text extracted from legal documents collected in raw HTML format. The same questions are used for both text representations, and results are reported separately for each corpus condition. For Zalo, reference text is retained when an extracted article is unavailable.

Second, a Bộ Công an (BCA) dataset is constructed from published public-service questions and legal citations in their answers. The evaluation examines whether the cited provisions are present in the corpus and identifies retrieval failures. The evaluation result is reported separately due to differences in data source from ALQAC and Zalo.

Third, the BGE-M3 embedding model is fine-tuned using legal question article pairs and evaluated in standalone and hybrid retrieval. BM25 and E5 remain unchanged, and fusion weights are held fixed when comparing pretrained and fine-tuned BGE-M3 within each hybrid. The analysis distinguishes development scores from results on test set queries.

1.3 Research Objectives

The main objective is to evaluate hybrid retrieval and fine-tuning of the BGE-M3 embedding model for Vietnamese legal information retrieval. The thesis has four specific objectives:

1. Compare BM25, pretrained BGE-M3, pretrained E5, and BM25+BGE-M3 using the same queries, corpus, and relevance annotations within each evaluation condition. BM25's evaluation scores serve as the baseline.

2. Determine whether adding E5 improves retrieval by comparing BM25+BGE-M3+E5 with BM25+BGE-M3.

3. Evaluate whether fine-tuning BGE-M3 improves standalone retrieval and whether any improvement is retained in BM25+BGE-M3 and BM25+BGE-M3+E5. Model selection uses development data, followed by evaluation on a separate test set. Fusion weights are held fixed when comparing pretrained and fine-tuned BGE-M3 within each hybrid.

4. Construct and evaluate the BCA collection from published public-service questions and legal citations in their answers. Assess the availability of cited provisions in the corpus, the accuracy of mappings between citations and document identifiers, and dependence among questions linked by shared legal references.

Retrieval effectiveness is measured primarily by NDCG@10. Hit@10, MRR@10, and MAP@10 provide additional measures of retrieval performance within the top ten results. Comparisons are reported separately for each dataset and corpus condition to establish the scope of any observed improvement.

1.4 Research Contributions

The thesis contribution is an empirical comparison of BM25, BGE-M3, E5, BM25+BGE-M3, and BM25+BGE-M3+E5 on ALQAC, Zalo, and BCA. The evaluation measures retrieval performance on datasets ALQAC, Zalo, and BCA. The results show that hybrid retrieval improves NDCG@10 over BM25 in most evaluated conditions, while adding E5 provides little to no additional benefit (Tables 5.3 and 5.4).

The second contribution is an evaluation of fine-tuning the BGE-M3 embedding model for standalone and hybrid retrieval. The first experiment selects the fine-tuned model based on hybrid retrieval performance on development data. Development scores improve, but evaluation on a test set does not demonstrate a hybrid improvement. The second experiment selects the model based on standalone dense retrieval performance on development data.

The result shows fine-tuning improves metric NDCG@10 but does not produce a meaningful improvement in either hybrid when fusion weights are the same (Tables 5.11 and 5.12).

The third contribution is the construction and evaluation of the BCA retrieval collection from published public-service questions and legal citations in their answers. The analysis examines the availability of cited provisions in the corpus and dependence among questions linked by shared legal references. The analysis also examines the sortfall failures of the BCA collection (Table 5.5).

1.5 Scope of the Thesis

The thesis evaluates the ranking of legal articles for Vietnamese questions on ALQAC, Zalo, and BCA. It compares BM25, BGE-M3, E5, and their hybrid configurations, and examines the effect of fine-tuning BGE-M3. The evaluation covers retrieval performance, corpus coverage, and the mapping of cited provisions to indexed documents. Answer generation, the quality of legal advice, user satisfaction, the currency of legal documents, and deployment performance are outside its scope.

The evaluation uses Vietnamese questions and legal texts. BGE-M3 and E5 are multilingual models, but the experiments do not assess retrieval in other languages or legal systems. Conclusions are limited to the evaluated datasets, corpus conditions, model configurations, and fusion weights. Applying the findings to other collections or downstream RAG systems requires further evaluation.

For BCA dataset the data is collected from publicly available questions and answers on the Ministry of Public Security's website on the date of collection. Many questions are refrencing private legal documents, which is inaccessible to the public. The evaluation does not attempt to access or retrieve any private legal documents, and the analysis is limited to publicly available information.

1.6 Thesis Organization

Chapter 1: Introduction presents Vietnamese legal information retrieval and the research gap addressed by the thesis. It states the research objectives, contributions, and scope.

Chapter 2: Literature Review reviews prior work on legal retrieval, BM25, multilingual embedding models, hybrid retrieval, and fine-tuning. It also discusses retrieval benchmarks and evaluation methods to establish the basis for the thesis's experimental comparisons.

Chapter 3: Theoretical Background explains lexical and dense retrieval, score normalization and fusion, and contrastive training. It defines relevance, evaluation metrics,

and the principles used to account for dependence among queries when interpreting results.

Chapter 4: Proposed Methodology describes dataset and corpus construction, including the BCA collection, and the retrieval pipeline. It specifies fusion weight selection, BGE-M3 fine-tuning, and the procedures for model selection and evaluation.

Chapter 5: Experimental Evaluation reports the retrieval comparisons, the effect of adding E5, and the effect of fine-tuning BGE-M3 and hybrid retrieval. It also analyzes BCA corpus coverage, dependence among questions linked by shared legal references, and retrieval failures.

Chapter 6: Conclusion and Future Work summarizes the findings and contributions in relation to the research objectives. It discusses the limitations of the evaluation and outlines directions for future work.

Chapter 2

Literature Review

This chapter reviews prior work on legal retrieval, multilingual embeddings, and hybrid fusion. It identifies the research gap addressed by this thesis and positions its contributions relative to the literature.

Figure 2.1 shows the position of retrieval in a Retrieval-Augmented Generation (RAG) system. The retrieval component searches the document sources and supplies information to the generator to support answer generation.

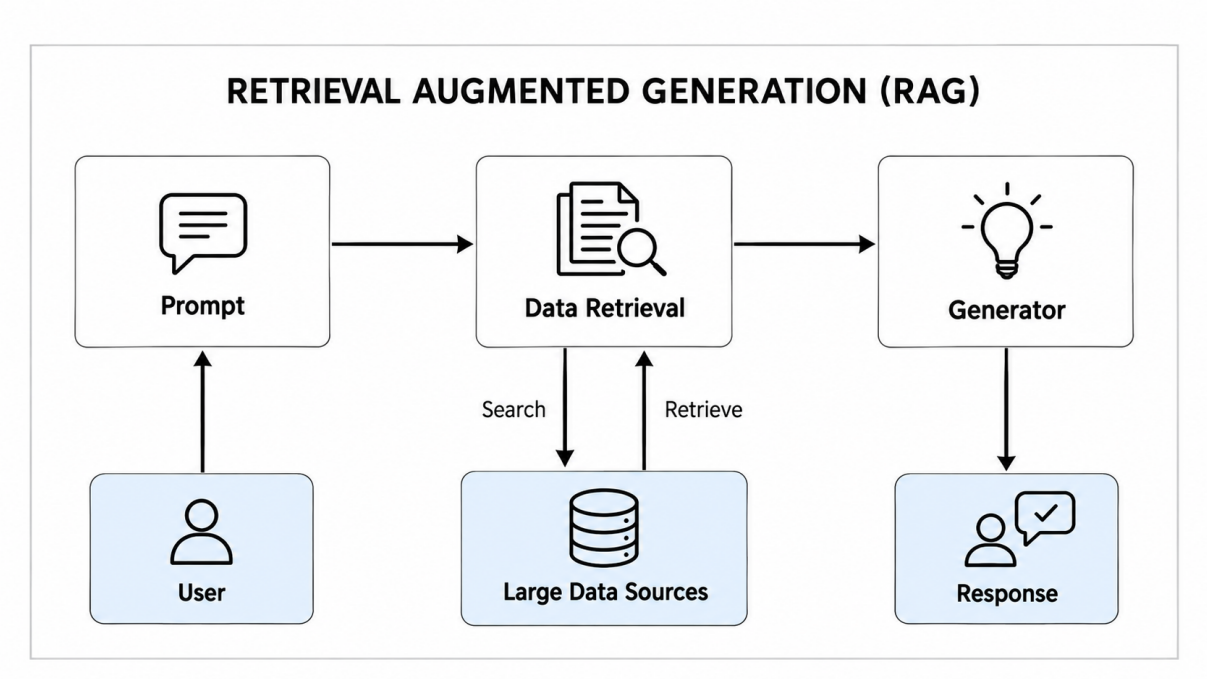


Figure 2.1: Position of retrieval in a Retrieval-Augmented Generation (RAG) system.

2.1 Legal Retrieval and Evidence

Legal information retrieval identifies and ranks legal materials relevant to a query. The retrieval unit may be a judgment, statute, article, or passage. The Competition on Legal Information Extraction and Entailment (COLIEE) separates retrieval tasks from entailment tasks, which assess whether legal texts support a given statement. Its case tasks use Canadian case law, while its statute tasks use Japanese bar-examination questions [7]. Comparisons with Vietnamese article retrieval must therefore account for differences in

the legal system, retrieval unit, relevance annotations, and evaluation metrics.

The Automated Legal Question Answering Competition (ALQAC) 2024 provides manually annotated Vietnamese legal questions and defines separate tasks for legal document retrieval and question answering. In the retrieval task, articles are relevant when they can be used to answer the question [8]. This thesis uses the available relevance annotations to assess how highly the recorded relevant articles are ranked.

Similarity in wording or topic alone does not establish relevance to a legal question. For example, a question about an application deadline may match an article concerning a different stage of the same procedure. In this hypothetical case, the article discusses the same topic but does not provide the required deadline. A relevant provision may also use different terms from those in the question. These possibilities motivate comparing lexical and dense retrieval using the same queries and relevance annotations.

The retrieval unit must also match the relevance annotations. An article may contain rules, exceptions, and references to other provisions, so splitting it into passages changes the context available to the retriever. When the annotations identify articles, retrieved passages must be mapped to their source articles before ranking performance is evaluated. Otherwise, passage retrieval and article retrieval would be assessed using different units.

Text preprocessing is another consideration in Vietnamese retrieval. PhoBERT uses word segmentation before subword tokenization and is evaluated on Vietnamese language-processing tasks [9]. Its preprocessing illustrates a model-specific requirement; its results do not establish that word segmentation improves the retrievers used in this thesis. For lexical retrieval, splitting text at spaces and segmenting it into words produce different vocabularies and term counts. For dense retrieval, preprocessing should follow the embedding model's documented requirements.

In the RAG workflow shown in Figure 2.1, retrieval supplies information to the generator. Lewis et al. combine a language model with retrieved passages for knowledge-intensive tasks [4]. This thesis evaluates the retrieval component by measuring article ranking against the available relevance annotations. Therefore, its results establish retrieval performance within those annotations, while any improvement in the accuracy of generated legal answers requires a separate evaluation.

2.2 Lexical Retrieval and BM25

Lexical retrieval ranks documents by matching query terms with indexed text. BM25 is a probabilistic ranking model that combines term frequency and document frequency with document length normalization [5]. Its scores therefore depend on both the terms in a document and their distribution in the indexed corpus.

A BM25 submission achieved second place in the legal case retrieval task of COLIEE 2021 [6]. This finding provides empirical support for using BM25 as a baseline in legal retrieval. Since that task concerns case law, its results do not establish BM25's effectiveness for Vietnamese legal article retrieval. The latter requires evaluation on the Vietnamese collections studied here.

This thesis compares dense and hybrid retrieval with BM25 using the same queries, corpus, and relevance annotations within each evaluation condition. BM25's evaluation scores serve as the baseline. Section 3.2 explains the scoring formula, and Section 4.4 specifies the preprocessing and retrieval implementation used in the experiments.

2.3 Dense Retrieval and Multilingual Embeddings

Dense retrieval represents queries and documents as embeddings and ranks documents by vector similarity. Sentence-BERT produces sentence embeddings that can be compared using cosine similarity [10]. Dense Passage Retrieval trains separate query and passage encoders for open-domain question answering [11]. Independent encoding allows document embeddings to be computed before a query is submitted and reused during retrieval.

BGE-M3 supports multilingual retrieval through dense, sparse, and multiple-vector representations [2]. Multilingual E5 provides embedding models in small, base, and large variants, trained using multilingual text pairs and labeled datasets [3]. This thesis evaluates BGE-M3's dense representations and multilingual-E5-base (E5). The results therefore apply to these configurations, rather than to every retrieval mode or model variant described in the source papers.

Single-vector retrieval represents each query and document with one vector. ColBERT instead retains contextualized token embeddings and computes relevance through interactions between query and document tokens after encoding [12]. This approach provides an alternative to single-vector retrieval. ColBERT is included here as related work and is not evaluated in the experiments.

Multilingual-E5-base is included to test whether it complements BGE-M3 in semantic retrieval. The models use different training procedures and embedding methods [2, 3]. The hypothesis is that these differences may produce complementary rankings, allowing E5 to improve retrieval for some questions whose relevant articles are ranked poorly by BGE-M3. Whether this improves hybrid retrieval is tested by comparing BM25+BGE-M3+E5 with BM25+BGE-M3.

The experiments also compare pretrained BGE-M3 and E5 with BM25 to assess standalone retrieval on the Vietnamese legal collections, and BM25+BGE-M3 with BM25 to assess hybrid retrieval. These comparisons distinguish the effect of combining lexical and dense retrieval from the effect of adding E5 to an existing hybrid.

2.4 Hybrid Retrieval and Score Fusion

Hybrid retrieval combines the outputs of different retrievers to produce a single ranking. Reciprocal Rank Fusion combines rankings using document positions rather than the original retrieval scores [13]. Weighted score fusion instead combines the component scores according to assigned weights. An analysis of lexical and semantic retrieval examines how score normalization and parameter selection affect fusion performance [14]. These findings motivate specifying both the normalization method and the procedure for selecting fusion weights.

Example of Score Fusion. Consider the legal question:

> *"Is an employee entitled to severance allowance after terminating an employment contract?"*

BM25 and BGE-M3 independently assign retrieval scores to candidate legal articles. Since their scores have different scales, each model's scores are normalized separately over the indexed corpus C. For model m, Min-Max normalization is defined as

$$s'_m(d) = \frac{s_m(d) - \min_{d' \in C} s_m(d')}{\max_{d' \in C} s_m(d') - \min_{d' \in C} s_m(d')},$$

assuming the maximum and minimum scores differ. The normalized scores are then combined using weights of 0.4 for BM25 and 0.6 for BGE-M3:

$$S(d) = 0.4 s'_{\text{BM25}}(d) + 0.6 s'_{\text{BGE-M3}}(d).$$

Table 2.1 shows hypothetical normalized scores for three candidate articles from the larger corpus. The article descriptions and weights are used only to illustrate the calculation.

Table 2.1: Illustrative BM25+BGE-M3 score fusion for three legal articles.

Legal article	**BM25**	**BGE-M3**	**Fused score**	**Final rank**
A: Severance allowance	0.90	0.95	**0.93**	**1**
B: Unemployment benefits	0.85	0.60	0.70	2
C: Contract termination	0.40	0.80	0.64	3

For Article A, the fused score is

$$S(A) = 0.4 \times 0.90 + 0.6 \times 0.95 = 0.93.$$

The articles are ranked by fused score in descending order. Article A ranks first because both models assign it high scores. The example illustrates how fusion combines lexical

matching from BM25 and embedding similarity from BGE-M3 into a single ranking score. It demonstrates the calculation, rather than establishing which provision legally answers the question.

The hypothesis is that lexical and dense retrieval may provide complementary rankings. A relevant article ranked poorly by one retriever may be ranked highly by another. Adding E5 to BM25+BGE-M3 tests whether another dense retriever improves a system that already combines lexical and semantic information. Its standalone performance does not establish that additional benefit.

This thesis uses normalized weighted score fusion and compares BM25+BGE-M3 with BM25+BGE-M3+E5. The weight search includes configurations that assign E5 a weight of zero, so adding E5 expands the available configurations without requiring its use. A higher score on the data used to select weights therefore does not establish an improvement on separate evaluation queries. The selected weights must be evaluated on queries excluded from weight selection.

The evaluation reports the effect of adding E5 separately for each dataset and corpus condition. Section 4.5 specifies score normalization and fusion, and Section 4.6 describes weight selection and evaluation. Reciprocal Rank Fusion is discussed as related work but is not evaluated in the experiments.

2.5 Fine-Tuning for Legal Retrieval and Hard Negatives

The thesis examines two fine-tuning experiments. The first selects the model based on hybrid retrieval performance on development data, followed by evaluation on a separate test set. The second selects the model based on standalone dense retrieval performance and then evaluates it in BM25+BGE-M3 and BM25+BGE-M3+E5. Section 4.7 describes the training procedures, and Section 5.6 reports the results and evaluation limitations.

Contrastive training teaches a retrieval model to give relevant passages higher similarity scores than passages labeled as irrelevant [11]. This thesis uses legal questions and articles to fine-tune BGE-M3 in this way.

Hard negatives are passages that a retriever ranks highly even though they are labeled as irrelevant. They help train the model to distinguish relevant passages from similar but incorrect matches. Experiments show that mixing hard negatives with randomly selected negatives, or selecting new hard negatives during training, can improve dense retrieval [16].

The hybrid comparisons replace pretrained BGE-M3 with fine-tuned BGE-M3 while keeping BM25, E5, and fusion weights unchanged. BGE-M3 is evaluated both on its own and within each hybrid. This tests whether fine-tuning improves standalone retrieval and

whether that improvement is retained when its scores are combined with those of the other retrievers.

2.6 Research Gap and Positioning

BGE-M3 and multilingual E5 provide pretrained embedding models for multilingual retrieval [2, 3]. Research on lexical and semantic score fusion examines the effects of normalization and parameter selection [14], while hard-negative training strategies have improved dense retrieval on benchmark datasets [16]. These studies provide a basis for evaluating multilingual embeddings, score fusion, and embedding model fine-tuning on Vietnamese legal collections.

The research gap addressed by this thesis is the evidence needed to assess these methods under the selected Vietnamese legal datasets and corpus conditions. The evaluation distinguishes improvements over BM25 from improvements due to adding E5 to BM25+BGE-M3. It also tests whether fine-tuning BGE-M3 improves standalone retrieval and whether any improvement is retained in the hybrid systems. Model selection on development data is reported separately from evaluation on a separate test set.

The comparisons use clean reference text and article text extracted from legal documents in raw HTML format.

The contribution is a controlled empirical comparison of established retrieval methods, together with the construction and evaluation of the BCA collection. The BCA analysis examines the availability of cited provisions and dependence among questions linked by shared legal references. Chapter 4 specifies the datasets and evaluation procedures, and Chapter 5 reports the results within those conditions.

2.7 Chapter Summary

This chapter reviewed legal retrieval, BM25, multilingual embeddings, score fusion, and fine-tuning with hard negatives. The reviewed studies motivate evaluating these methods on Vietnamese legal collections, with separate comparisons for improvements over BM25, the contribution of E5, and the effect of BGE-M3 fine-tuning in standalone and hybrid retrieval. Chapter 3 presents the theoretical background and evaluation metrics used in these comparisons.

Chapter 3

Theoretical Background

This chapter presents the theoretical foundations of retrieval models, score fusion, model training, and evaluation metrics used in the thesis. Chapter 4 describes how these methods are implemented and evaluated.

3.1 Queries, Articles, and Relevance

Information retrieval ranks indexed documents according to their relevance to a query [1]. In this thesis, the query is a Vietnamese legal question, and the retrieval unit is a legal article.

The corpus contains laws (Luật), decrees (Nghị định), and circulars (Thông tư). A document may be organized into parts (phần), chapters (chương), sections (mục), subsections (tiểu mục), articles (điều), clauses (khoản), and points (điểm), although not every document uses all these levels [21]. Each article belongs to one legal document and may contain several clauses and points.

Chunking divides document text into units for indexing and retrieval. The choice of unit can affect retrieval performance [22]. This thesis uses articles as the retrieval units. For documents collected in raw HTML format, the extracted text is separated at article boundaries. Each article record contains its text, document identifier, and article identifier. Clauses and points are kept within their source article. Section 4.3 describes corpus construction and the Zalo reference-text fallback when extraction is unavailable.

Chunking can affect recall because it determines the content available to the retriever before encoding. If a chunk incorrectly combines three articles, one vector represents their combined content. This may make a specific provision harder to retrieve and associate text with the wrong article identifier. Improving the encoder does not repair missing text or incorrect identifiers in the corpus. Article boundaries and the match between text and identifiers must therefore be checked when investigating retrieval errors.

Example of an article-level record. The following example contains two article excerpts from the supplied sample for Decree No. 340/2025/NĐ-CP. The title and article texts are shortened for display, with ellipses marking omitted text. The example illustrates the record format; the complete sample retains each article's clauses and points.

```
{
  "law_id": "340/2025/NĐ-CP",
  "title": "NGHỊ ĐỊNH SỐ 340/2025/NĐ-CP ...",
  "url": "https://vbpl.vn/van-ban/chi-tiet/nghi-dinh-so-340-2025-nd-cp-quy-dinh-ve-xu
  ↪  -phat-vi-pham-hanh-chinh-trong-linh-vuc-tien-te-va-ngan-hang--187101",
  "articles": [
    {
      "id": "1",
      "text": "Điều 1. Phạm vi điều chỉnh\n1. Nghị định này quy định các hành vi vi
      ↪  phạm hành chính ... trong lĩnh vực tiền tệ và ngân hàng.\n2. Vi phạm hành
      ↪  chính trong lĩnh vực tiền tệ và ngân hàng bao gồm: ..."
    },
    {
      "id": "2",
      "text": "Điều 2. Đối tượng bị xử phạt vi phạm hành chính\n1. Nghị định này áp
      ↪  dụng đối với tổ chức, cá nhân có hành vi vi phạm hành chính trong lĩnh vực
      ↪  tiền tệ và ngân hàng.\n2. Tổ chức quy định tại khoản 1 Điều này gồm: ..."
    }
  ]
}
```

The fields `law_id`, `title`, and `url` describe the legal document. Each entry in `articles` is a separate chunk with its own `id` and `text`. Article 1 and Article 2 therefore receive separate embeddings. Together, `law_id` and the article `id` identify each retrieval unit.

Relevance targets identify the documents or articles recorded as relevant to each query. ALQAC and Zalo use the available dataset annotations. For BCA, the question is used as the query, and citations in its published answer are extracted to form relevance targets. Document names and article numbers are normalized and matched to corpus identifiers. The answer text is used for target extraction and is not supplied to the retriever. Table 4.2 illustrates this process.

An article-level target specifies a document and an article within it. A law-level target specifies only the document. In the dataset, "law-level" also covers decrees and circulars. Duplicate targets are removed, and each target receives evaluation credit at most once. Retrieving several articles from one document therefore does not multiply credit for a single law-level target. Section 3.6 defines the matching rules and evaluation metrics.

Retrieval returns a list of article identifiers ordered by retrieval score. These identifiers link the results to the article text and source document. A RAG generator can use the retrieved text to produce an answer, while this thesis evaluates the article ranking. Corpus coverage measures target availability, and ranking performance measures target positions.

3.2 Lexical Scoring with BM25

3.2.1 Term Frequency and Inverse Document Frequency

Term frequency and inverse document frequency (TF-IDF) are combined to weight indexed terms. A term receives a higher weight when it occurs frequently in an article but appears in relatively few documents across the corpus [1]. This illustrates how lexical weighting combines term occurrence with term rarity. The experiments use BM25 as the lexical baseline; TF-IDF is included here as background.

3.2.2 BM25

BM25 is a popular lexical scoring function that combines term frequency, inverse document frequency, and document length normalization. For a query q and an article d, a common scoring formula sums the contributions of query terms t [5]:

$$\mathrm{BM25}(q,d) = \sum_{t \in q} \mathrm{IDF}(t) \frac{f(t,d)(k_1+1)}{f(t,d) + k_1\left(1 - b + b|d|/\,\mathrm{avgdl}\right)}.$$

The quantities in the formula are:

- $f(t,d)$: the number of occurrences of term t in article d.
- $|d|$: the article length, measured in indexed tokens.
- avgdl: the average article length in the indexed corpus.
- $\mathrm{IDF}(t)$: the inverse document frequency weight of term t.

The parameter k_1 controls term frequency saturation: repeated occurrences of a term contribute progressively less to its score. The parameter b, between zero and one, controls document length normalization. Setting $b = 0$ removes length normalization, while $b = 1$ applies full length normalization [1].

BM25 scores depend on the indexed text. Text extraction can change term counts, article lengths, and corpus statistics, so different corpus representations may produce different scores. A query term absent from an article contributes zero to its BM25 score. Section 4.4 describes the lexical preprocessing used in the experiments.

BM25 relies on matching indexed terms. A relevant article expressed using different terms from the query may therefore receive a low score. The thesis compares BM25 with dense and hybrid retrieval to evaluate whether embedding similarity improves the ranking. BM25 scores determine article order, while evaluation metrics such as NDCG@10 measure ranking quality against the relevance annotations. The baseline is BM25's evaluation performance on the target collections.

3.3 Dense Retrieval and Vector Similarity

An embedding is a numerical vector that represents text. In a bi-encoder architecture, the query and article are encoded separately. Dense retrieval then ranks articles by the similarity between their embeddings and the query embedding. Sentence-BERT produces sentence embeddings for similarity comparison [10]. Dense Passage Retrieval trains query and passage encoders to retrieve relevant passages [11]. Article embeddings can be computed in advance and reused across queries.

Figure 3.1 illustrates separate query and document encoding, similarity scoring, and ranking. The documents in this workflow correspond to legal articles in this thesis.

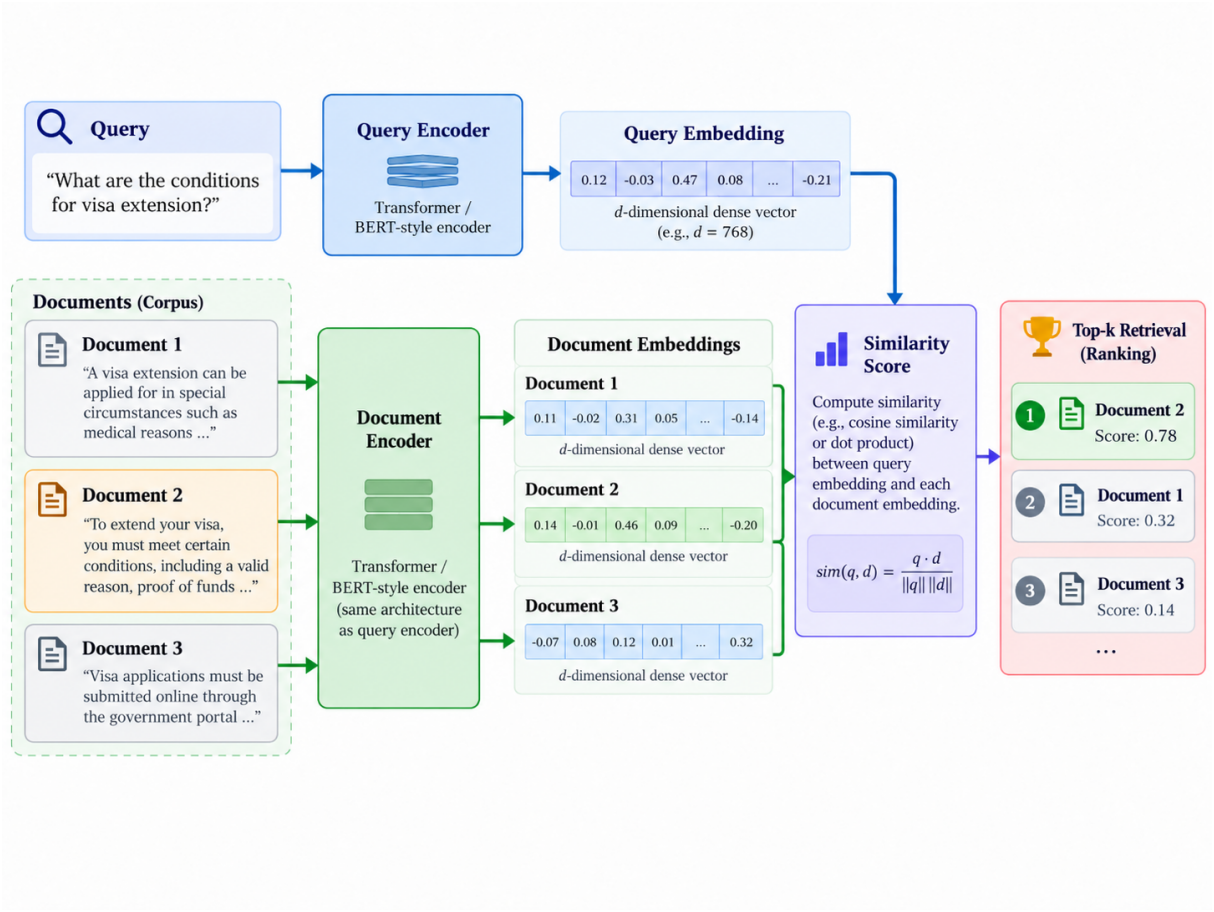


Figure 3.1: Dense retrieval workflow. The displayed vectors and scores are illustrative and do not represent experimental results.

For a query q and an article d, u and v are their respective embedding vectors. Two common similarity measures are the dot product and cosine similarity:

$$s_{\text{dot}}(q, d) = u^\top v, \qquad s_{\cos}(q, d) = \frac{u^\top v}{\|u\|_2 \|v\|_2}.$$

The dot product $u^\top v$ sums the products of corresponding vector components. The

quantities $\|u\|_2$ and $\|v\|_2$ are the vector lengths. Cosine similarity divides the dot product by these lengths and is defined for nonzero vectors. When both vectors are normalized to length one, the dot product equals cosine similarity.

A pooling operation produces one embedding from the model's token representations. Input longer than the selected token limit is truncated, so the embedding may omit content that remains in the stored article. ColBERT retains multiple token embeddings for matching [12]. This thesis uses single-vector dense retrieval.

3.3.1 BGE-M3

BGE-M3 has three main features:

1. **Multi-Functionality**

   BGE-M3 supports three retrieval methods [24]:

   - **Dense Retrieval:** Uses dense vector representations to capture semantic meaning.
   - **Sparse Retrieval:** Uses sparse representations to match important keywords and lexical features.
   - **Multi-Vector Retrieval:** Uses multiple vectors with late interaction mechanisms, similar to ColBERT, to capture relevance through comparisons between query and document tokens.

2. **Multi-Linguality**

   - Supports more than 100 languages [2].
   - Enables cross-lingual retrieval, allowing a query to retrieve documents written in a different language [2].

3. **Multi-Granularity**

   - Handles text inputs of varying lengths, from short sentences to long documents [24].
   - Supports input sequences of up to 8,192 tokens for encoding short passages and longer documents [24].

This thesis uses BGE-M3's dense representations for retrieval and fine-tuning. The baseline experiments use a maximum input length of 512 tokens (Table 4.3). The hybrid systems combine BM25 scores with BGE-M3 dense similarity scores. BGE-M3's sparse and multiple-vector modes are outside the evaluated configurations.

3.3.2 Multilingual-E5-base

Multilingual E5 is pretrained on multilingual text pairs and then fine-tuned using labeled datasets [3]. This thesis uses `intfloat/multilingual-e5-base`, which produces vectors with 768 components. It uses mean pooling to average token representations, excluding padding tokens. The documented retrieval setup normalizes the vectors and limits input to 512 tokens [23].

E5 requires the prefix `query:` for questions and `passage:` for article text, including Vietnamese inputs [23]. These prefixes specify the role of each input during encoding.

E5 is not fine-tuned in this thesis. Its contribution is measured by comparing BM25+BGE-M3+E5 with BM25+BGE-M3.

3.4 Normalization and Fusion

3.4.1 Lexical–Semantic Hybrid Retrieval

Hybrid retrieval combines lexical and semantic evidence. Its central hypothesis is that the combined ranking improves on its components, but effectiveness depends on the fusion configuration [14].

Full-corpus fusion scores every document with every channel. Candidate-union fusion instead ranks only documents returned by first-stage retrievers; weights cannot recover a target excluded from every candidate list. Candidate depth and score combination are therefore separate design choices [14].

3.4.2 Multi-Retriever Fusion

For normalized BM25, BGE-M3, and E5-base scores $\widetilde{S}_b$, $\widetilde{S}_g$, and $\widetilde{S}_e$, three-way fusion is

$$S(q,d) = w_b\widetilde{S}_b(q,d) + w_g\widetilde{S}_g(q,d) + w_e\widetilde{S}_e(q,d), \quad w_b, w_g, w_e \geq 0, \quad w_b + w_g + w_e = 1.$$

Setting $w_e = 0$ recovers BM25+BGE-M3 when all other settings are unchanged. A search containing this boundary cannot have a lower optimal development score than its two-way subset, but improvement on unseen data is not guaranteed. A selected zero E5 weight indicates that the additional channel is unused.

Complementarity may affect early precision, first-hit position, or target recovery differently. Multiple ranking metrics are therefore needed; positive fusion weights alone do not establish useful complementary errors.

3.4.3 Score Normalization and Fusion

For raw channel score S_j and normalization candidate set C_q, min–max normalization is [14]

$$\widetilde{S}_j(q,d) = \frac{S_j(q,d) - a_j(q)}{b_j(q) - a_j(q)}, \quad a_j(q) = \min_{x \in C_q} S_j(q,x), \quad b_j(q) = \max_{x \in C_q} S_j(q,x).$$

The expression requires $b_j(q) > a_j(q)$. Constant-score channels require a deterministic rule, such as mapping all values to zero, because they provide no within-query ordering evidence.

An alternative is z-score normalization,

$$Z_j(q,d) = \frac{S_j(q,d) - \mu_j(q)}{\sigma_j(q)},$$

where $\mu_j(q)$ and $\sigma_j(q) > 0$ are the mean and standard deviation over C_q. Unlike min–max scores, z-scores are not bounded by zero and one. Both methods depend on the candidate population: adding documents can change normalized values without changing existing raw scores.

Reciprocal rank fusion (RRF) combines positions instead of score magnitudes [13]:

$$\mathrm{RRF}(d) = \sum_{j \in J_d} \frac{1}{c + r_j(d)}.$$

Here, $r_j(d)$ is a one-based rank, J_d contains the rankings that include d, and $c > 0$ is a smoothing constant. RRF avoids direct score-scale comparison but remains sensitive to rank depth and parameter choices. Evidence favoring tuned convex fusion in particular settings does not establish a universal ordering; normalization, weights, candidate depth, and selection data must all be specified [14].

3.5 Contrastive Adaptation and Negative Sampling

Legal-domain language-model pretraining and supervised retrieval fine-tuning are different interventions. LEGAL-BERT adapts language representations to legal text, whereas DPR learns query–passage ranking from relevance supervision. Familiarity with legal prose does not itself establish retrieval effectiveness [15, 11].

3.5.1 Contrastive Learning

A single-positive InfoNCE-style loss encourages a relevant document to outscore competing documents [19, 11]:

$$\mathcal{L}(q) = -\log \frac{\exp(s(q, d^+)/\tau)}{\exp(s(q, d^+)/\tau) + \sum_{d^- \in N(q)} \exp(s(q, d^-)/\tau)}.$$

Here, d^+ is a labeled positive, $N(q)$ is the negative set, s is similarity, and $\tau > 0$ is temperature. Temperature changes the emphasis on score differences, while batch composition determines the competitors. Lower training loss does not guarantee better retrieval on unseen data.

For multiple positives, a set-positive loss can sum positive exponentials in the numerator. Alternatively, positive-specific losses can be averaged while masking other known positives from negative positions. These objectives are not interchangeable; Chapter 4 states the loss and masking policy of each study.

3.5.2 Negative Sampling

Random negatives are sampled without a high-rank requirement; in-batch negatives reuse passages from other examples; hard negatives are high-ranking candidates without positive labels [11, 16].

Excluding known positives prevents label contradictions but does not prove that unjudged articles are irrelevant. In particular, selecting one positive must not cause another known positive to become an unmasked negative. Explicit-negative filtering, in-batch masking, and semantic review provide different safeguards and should be reported separately.

3.5.3 Hard-Negative Mining

Static mining fixes competitors before training; dynamic mining refreshes them as the model changes [16]. Reproduction requires the miner checkpoint, corpus snapshot, ranking range, filtering rules, and any refresh schedule.

Adaptation must be evaluated both alone and within the hybrid: a dense improvement may overlap with BM25's strengths or reduce useful disagreement. Retaining pretrained fusion weights tests component substitution, whereas retuning on permissible development data tests adaptation of the complete system.

3.6 Target Matching and Evaluation Metrics

The evaluator selects ten candidates by partial partitioning and sorts them by decreasing score. Each retrieved article is checked against unmatched targets. An empty article identifier or `toanvan` denotes a law-level wildcard; otherwise, both law and article identifiers must match. Each target earns credit at most once, so retrieving several articles from one law does not multiply credit for a single whole-law target.

In the following formulas, Q is the set of evaluation queries, and R_q contains the distinct recorded relevance targets for query q. The indicator $y_{q,r}$ equals one if the article at rank r matches a previously unmatched target, and zero otherwise. The primary metric is [18]

$$\mathrm{NDCG@10}(q) = \frac{\sum_{r=1}^{10} y_{q,r}/\log_2(r+1)}{\sum_{r=1}^{\min(|R_q|,10)} 1/\log_2(r+1)}.$$

Queries without targets receive zero. Unavailable targets remain in the ideal-gain denominator, making coverage part of end-to-end effectiveness. Mixed law-level and article-level targets remain subject to the implemented matching order.

The hit metric is

$$\mathrm{Hit@}k(q) = \mathbb{1}\left[\sum_{r=1}^{k} y_{q,r} > 0\right],$$

reported at $k = 1, 5, 10$. It indicates whether any target is recovered, not the proportion recovered. True target recall is $\mathrm{Recall@}k(q) = \sum_{r=1}^{k} y_{q,r}/|R_q|$ for nonempty R_q. Hit and recall coincide for single-target queries but can differ for multi-target queries; the canonical baseline aggregate does not report true target recall, including Recall@100 [1].

With r_q^* denoting the first matching rank,

$$\mathrm{MRR@10} = \frac{1}{|Q|}\sum_{q\in Q}\frac{\mathbf{1}[r_q^* \le 10]}{r_q^*},$$

where a missing match contributes zero. The implemented average-precision metrics are

$$P_q(r) = \frac{1}{r}\sum_{j=1}^{r} y_{q,j},$$
$$\mathrm{AP@10}(q) = \frac{\sum_{r=1}^{10} y_{q,r} P_q(r)}{\max(1, \min(|R_q|, 10))},$$
$$\mathrm{MAP@10} = \frac{1}{|Q|}\sum_{q\in Q}\mathrm{AP@10}(q).$$

This truncated AP denominator may differ from other implementations. All aggregate metrics are query-macro means, including zero-score queries, unless an answerable-only cohort is explicitly stated. They measure annotated target recovery, not exhaustive legal

relevance [1].

3.7 Grouped Evaluation and Statistical Interpretation

Queries can share target laws, and multi-law queries connect those laws into components. Keeping each component within one fold prevents reference-law overlap across fusion selection and evaluation. It does not partition the searchable corpus or establish that encoders have never seen the legal text.

Query-, component-, and fold-macro averages differ when group sizes differ. Confidence intervals must identify their resampling unit; variation across training seeds is not uncertainty over a new query population. Likewise, an interval spanning zero neither proves equivalence nor erases an observed difference. Chapter 5 interprets the stored statistical tests within these boundaries [20].

3.8 Chapter Summary

BM25 and bi-encoders supply lexical and semantic scores; fusion combines them, and contrastive adaptation changes the dense representation. Target-aware metrics and grouped evaluation assess these interventions under explicit corpus and selection boundaries.

Chapter 4

Proposed Methodology

This chapter defines the retrieval task, datasets, scoring methods, fusion procedure, evaluation design, and legal-domain adaptation studies. The method uses established retrieval models and a controlled evaluation pipeline; the contribution lies in the dataset construction, experimental comparison, and interpretation of results rather than in a new ranking algorithm.

4.1 Problem Formulation

Let $D = \{d_1, \ldots, d_n\}$ denote an ordered corpus of legal retrieval units. Each unit contains article text and a normalized legal identifier. For a query q, the relevance set R_q contains the law–article targets recorded for that question. A retrieval configuration θ produces a ranked list

$$\pi_q(\theta) = (d_{(1)}, d_{(2)}, \ldots, d_{(k)}),$$

where $k = 10$ for the primary evaluation window.

The selection objective is mean NDCG@10 on a designated selection population:

$$\theta^* = \arg\max_{\theta \in \Theta} \frac{1}{|Q_{\text{sel}}|} \sum_{q \in Q_{\text{sel}}} \text{NDCG@10}(\pi_q(\theta), R_q).$$

The candidate set Θ contains tested fusion weights and, in the adaptation studies, selected encoder-training recipes. Labels are used only for selection and evaluation. At retrieval time, the system receives the query text and the indexed corpus; it does not receive the answer, target labels, or relevance judgments.

This formulation makes two limitations explicit. First, a target absent from the searchable corpus cannot be retrieved by any ranking model. Second, retrieval effectiveness is not the same as generated legal-answer correctness. The study therefore evaluates article ranking, not downstream legal advice.

4.2 Retrieval and Evaluation Pipeline

The baseline architecture compares individual retrievers and weighted hybrids. BM25 provides the lexical score. BGE-M3 and multilingual-E5-base provide dense vector scores. The two-way hybrid combines BM25 and BGE-M3; the three-way hybrid combines BM25, BGE-M3, and E5.

Each component scores the same ordered corpus. Scores are normalized per query and combined by weighted addition. The resulting top-ten rankings are evaluated against the recorded targets. The system does not use a cross-encoder reranker, query router, query expansion module, approximate nearest-neighbor filter, or answer-generation stage. This restriction keeps the experiment focused on first-stage lexical–semantic retrieval.

Figure 4.1 summarizes the baseline scoring and evaluation architecture.

4.3 Datasets and Corpus Construction

The evaluation covers ALQAC, Zalo, and BCA. ALQAC and Zalo are each evaluated in two corpus representations: a clean reference representation and a parsed-text representation. The paired A/B conditions share query populations and therefore should not be treated as independent datasets. BCA is evaluated separately because its target graph contains strong dependency between questions that cite the same laws.

Table 4.1: Canonical baseline evaluation conditions.

Condition	**Queries**	**Articles**	**Evaluation boundary**
ALQAC A, clean	811	3,363	Law-component grouped out-of-fold evaluation over the clean reference representation
ALQAC B, parsed	811	3,381	Same ordered queries with separately constructed parsed articles
Zalo A, clean	3,196	61,425	Law-component grouped out-of-fold evaluation over the clean reference representation
Zalo B, parsed with fallback	3,196	61,425	Same ordered queries with parsed text where available and reference text retained otherwise

Continued on next page

Condition	Queries	Articles	Evaluation boundary
BCA, canonical evaluation	546	14,789	Giant-component holdout plus small-component out-of-fold evaluation

In ALQAC A, reference legal articles are flattened while preserving article identity. ALQAC B uses parsed source material; parsing changes article boundaries and can change the number of retrieval units. In Zalo A, reference articles define the clean corpus. In Zalo B, parsed text replaces reference text where an aligned parsed version is available, while fallback preserves article availability when parsing fails. Consequently, the Zalo A/B comparison evaluates a specific parsed-with-fallback construction, not the isolated effect of parsing alone.

The BCA collection is constructed from public questions and published answers associated with the Bộ Công an portal. The questions supply retrieval inputs. The published answers provide citation evidence from which legal targets are derived. The resulting collection contains 652 source records, of which 546 are retained for retrieval evaluation because they have at least one extracted law reference. Later coverage analysis distinguishes answerable, partially answerable, and unavailable target cases.

4.3.1 BCA Target Construction

The BCA construction process converts public question-answer records into a retrieval dataset. Question text is cleaned and retained as the query. Answer text is used only to extract law and article references; it is not appended to the query at retrieval time. Extracted law names are normalized and aligned to legal identifiers in the searchable corpus. Article identifiers are normalized separately so that matching remains identity-based rather than text-similarity-based.

Figure 4.2 summarizes the construction process. The process is useful because it produces naturally occurring legal information needs outside the competition collections. Its limitations are equally important: published answers are not exhaustive relevance judgments, citation extraction is heuristic, and some cited material is absent from the indexed corpus.

4.3.2 Worked BCA Example

Table 4.2 illustrates how one public question becomes a retrieval instance. The question asks whether apartment buildings are subject to mandatory fire-prevention equipment requirements. The published answer cites a legal instrument and relevant articles. The dataset stores the question as the query and the cited provisions as target identities.

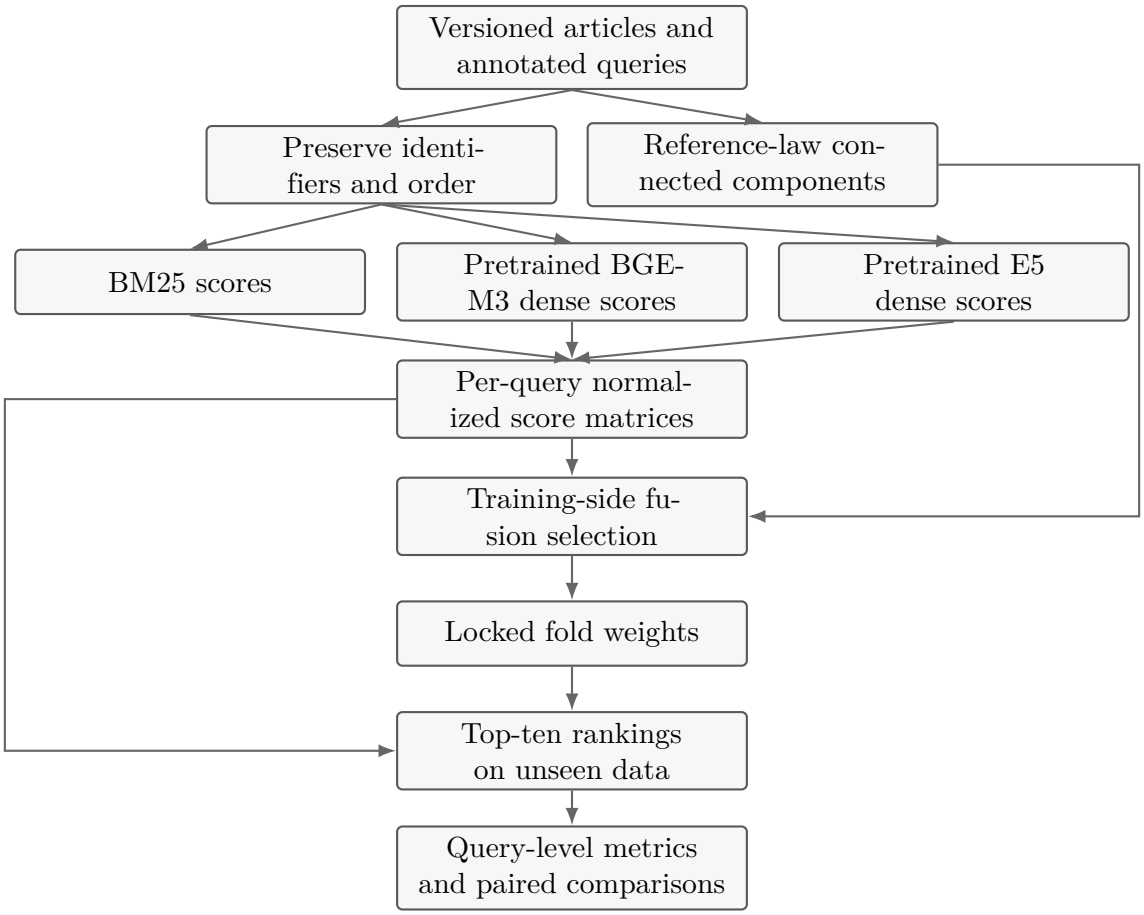


Figure 4.1: Baseline scoring and evaluation architecture.

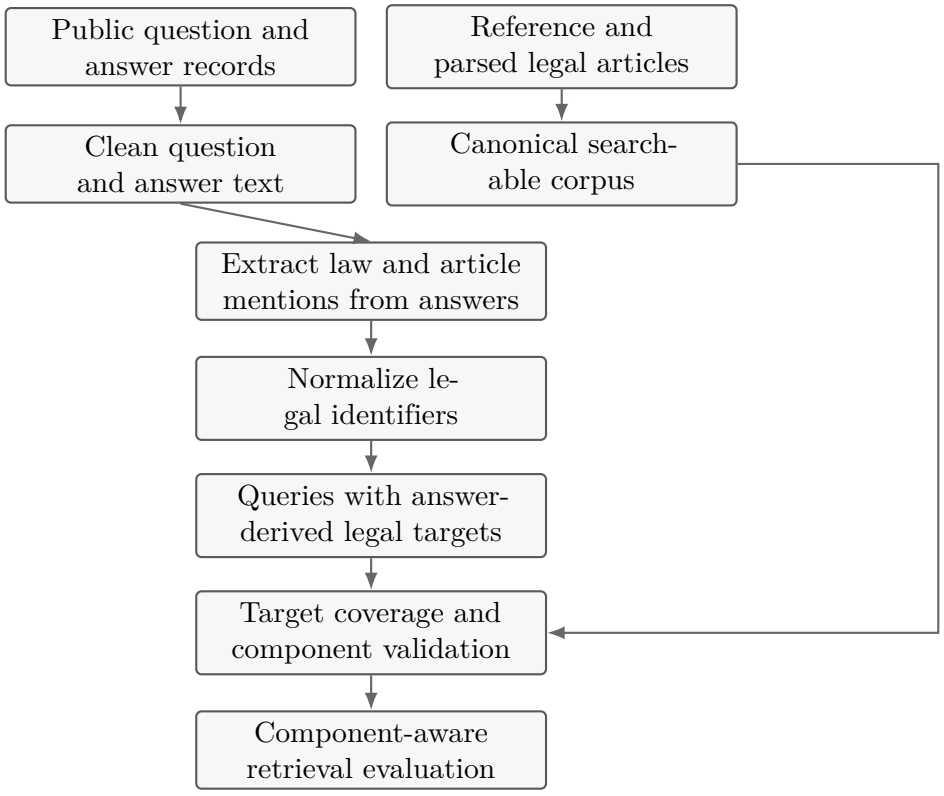


Figure 4.2: Construction of the BCA retrieval collection.

Table 4.2: Worked BCA question-to-target transformation.

Stage	**Representation**	**Interpretation**
Stored query	Vietnamese question about mandatory fire-prevention equipment for apartment buildings	Query text supplies the retrieval input
Extracted answer references	A cited decree with whole-law and article-level mentions	Published answer citations provide target candidates
Normalized law identity	Standardized law identifier for the cited decree	Citation text is aligned to the corpus identity space
Canonical target identity	Normalized law identifier plus article identifier	Evaluation matches ranked documents by legal identity
Retriever input	Query text and candidate article text	The published answer is not used as a query-time feature

The example demonstrates the collection design but not legal completeness. A published answer may omit useful supporting provisions, and an extracted citation may require legal review before being treated as an exhaustive relevance judgment.

4.4 Lexical and Dense Representations

BM25 scores are computed from tokenized query and article text. The lexical preprocessing lowercases text, removes non-word punctuation as token separators, and splits on whitespace. This simple policy is treated as part of the fixed lexical baseline. The experiments do not claim to evaluate every possible Vietnamese lexical analyzer or tokenization strategy.

Dense scores are produced by pretrained BGE-M3 and multilingual-E5-base encoders. Query and article embeddings are L2-normalized, and dense similarity is the vector dot product:

$$s_{\text{dense}}(q, d) = \frac{\mathbf{u}_q^\top \mathbf{v}_d}{\max(\|\mathbf{u}_q\|_2, 10^{-10}) \max(\|\mathbf{v}_d\|_2, 10^{-10})}.$$

Table 4.3: Baseline dense-model identity.

Property	**BGE-M3**	**Multilingual-E5-base**
Model family	BGE-M3 multilingual embedding model	E5 multilingual text-embedding model
Embedding dimension	1,024	768
Experimental role	Primary pretrained dense component	Additional pretrained dense component
Baseline maximum input length	512 tokens	512 tokens
Adaptation status	Adapted in the fine-tuning studies	Frozen throughout the thesis

The evaluation uses exhaustive corpus scoring rather than an approximate nearest-neighbor index. Therefore, a mapped and available target outside the top ten reflects scoring and fusion behavior, not candidate-generation recall. This design simplifies scientific interpretation, although it does not establish production latency.

4.5 Score Normalization and Fusion

BM25 and dense scores have different numeric scales, so they are normalized before fusion. For retriever r, raw score $s_r(q, d)$, and corpus C, the implemented per-query min–max normalization is

$$\widetilde{s}_r(q, d) = \frac{s_r(q, d) - \min_{d' \in C} s_r(q, d')}{\max\left(\max_{d' \in C} s_r(q, d') - \min_{d' \in C} s_r(q, d'), \epsilon\right)}.$$

The small constant ϵ prevents division by zero. Normalization is applied independently for each query and channel over the full corpus.

The two-way hybrid score is

$$S_2(q, d) = w_b \widetilde{s}_b(q, d) + (1 - w_b)\widetilde{s}_g(q, d),$$

where b denotes BM25 and g denotes BGE-M3. The three-way score is

$$S_3(q, d) = w_b \widetilde{s}_b(q, d) + w_g \widetilde{s}_g(q, d) + w_e \widetilde{s}_e(q, d), \quad w_b + w_g + w_e = 1,$$

where e denotes E5. Weights are nonnegative. The evaluated grid uses increments of 0.1. Thus, the two-way grid contains 11 candidate settings and the three-way simplex contains

66 candidate settings, including zero-weight boundaries.

For each split, training-side mean NDCG@10 selects the fusion weights, which are then fixed for evaluation on unseen data. Because the three-way grid contains the two-way boundary, a higher training-side fit does not by itself prove a benefit on unseen data. The two-way versus three-way comparison on unseen data in Chapter 5 is therefore the relevant evidence for E5's incremental value.

4.6 Evaluation Design

The baseline evaluation uses law-component grouping to reduce leakage from shared reference laws. Query groups are assigned to folds so that reference-law components are not split between training-side fusion selection and evaluation on unseen data. The resulting out-of-fold predictions measure a selection procedure rather than one universal global weight vector.

Algorithm 4.1 Grouped selection of fusion weights and out-of-fold evaluation

Require: Ordered queries Q, corpus D, aligned score matrices, grouped folds, candidate weight grid W
Ensure: Selected weights per fold and query metrics on unseen data
for each fold f **do**
    Let T be the training-side query indices and V the unseen-data indices
    for each candidate weight vector $w \in W$ **do**
        Rank D for every query in T using the weighted score sum
        Compute mean query NDCG@10
    end for
    Select the first candidate with the highest training-side mean
    Rank all queries in V using the selected weights
    Store query-level metrics on unseen data
end for
Aggregate unseen-data query rows across folds

Algorithm 4.2 Article ranking with a locked hybrid configuration

Require: Query q, corpus D, retriever scores, locked fusion weights w
Ensure: Ten highest-ranked articles
Normalize each retriever's scores over the full corpus for q
for each article $d \in D$ **do**
    $\text{combined}[d] \leftarrow \sum_r w_r \, \text{normalized_score}[r, d]$
end for
return the ten articles with highest combined score

For BCA, ordinary query-level evaluation would overstate reliability because many questions are connected through shared legal targets. The corrected protocol distinguishes a giant-component holdout, small-component out-of-fold evaluation, and a combined descriptive view. Chapter 5 reports all three because they answer different questions: performance on the dominant connected component, performance across smaller components, and descriptive behavior over the entire retained BCA collection.

4.7 Legal-Domain Fine-Tuning

Fine-tuning tests whether legal-domain supervision improves BGE-M3. The adaptation studies change only the BGE-M3 dense component. BM25 and E5 remain fixed. The primary hybrid evaluation then replaces pretrained BGE-M3 with the adapted BGE-M3 while keeping the relevant fusion policy explicit. This thesis refers to the two experiments as Study A: Hybrid-Oriented Adaptation and Study B: Dense-Oriented Adaptation.

4.7.1 Training Pairs and Negatives

Study A: Hybrid-Oriented Adaptation uses Zalo query–article pairs. For each eligible query, the training builder selects a known relevant article available in both clean and parsed views, creating paired training examples from the two corpus representations. Negative passages are mined from lexical and dense rankings after excluding known positives, duplicates, unavailable documents, and laws reserved for evaluation on unseen data.

Table 4.4: Training groups and protected folds in the completed hybrid fine-tuning study.

Stage	Train folds	Queries	A/B groups	Protected folds	Protected laws	Law overlap
Pilot	2, 3, 4	1,917	3,834	0, 1	258	0
Final training	1, 2, 3, 4	2,556	5,112	0	110	0

The zero-overlap checks protect the law identifiers reserved for evaluation on unseen data from appearing in the constructed training groups. They do not prove absence of shared legal concepts or similar language, and they do not make the final evaluation a test on unseen legal systems.

4.7.2 Training Objective and Configuration

Study A trains BGE-M3 with a single-positive in-batch contrastive objective. Each query group contains one selected positive and seven explicit negatives. With batch size B, group size $G = 8$, positive passage index $y_i = iG$, normalized query vector $\mathbf{q}_i$, normalized passage vector $\mathbf{p}_j$, and temperature $\tau = 0.02$, the implemented objective is

$$\mathcal{L}_{\text{batch}} = -\frac{1}{B}\sum_{i=0}^{B-1} \log \frac{\exp(\mathbf{q}_i^\top \mathbf{p}_{y_i}/\tau)}{\sum_{j=0}^{BG-1} \exp(\mathbf{q}_i^\top \mathbf{p}_j/\tau)}.$$

Table 4.5: Locked configuration for the completed final-training runs.

Parameter	**Verified value**
Base encoder	BGE-M3
Adaptation	Full dense encoder
Epochs	2
Learning rate	2.0000×10^{-5}
Contrastive query batch	4
Gradient accumulation	8
Optimizer effective batch	32 query groups
Passage group	1 selected positive and 7 explicit negatives
Query / passage maximum length	256 / 512 tokens
Temperature	0.0200
Pooling / normalization	First-token hidden state / L2 normalization
Seeds	42, 43, 44
Frozen E5 component	Multilingual-E5-base, unchanged throughout adaptation

Pilot training uses folds 2–4; fold 1 selects the learning rate, epoch, and contrastive batch setting by hybrid NDCG@10. After the recipe is locked, final models are retrained on folds 1–4 using three seeds. Fold 0 is reserved for Study A's corrected hybrid comparison on data unseen during training and selection. Because a flawed preliminary analysis had already exposed fold 0, the corrected result is treated as a post-hoc evaluation on unseen data, not as a pristine first-use test.

Figure 4.3 summarizes the selection and evaluation sequence.

4.7.3 Study B: Dense-Oriented Adaptation

Study B evaluates BGE-M3 as a dense retriever before integrating it into hybrids. This study uses a separate recipe with an all-positive contrastive mask, five explicit negatives, a larger true contrastive batch requirement, and a Zalo-only first passing data stage. Its selected setting is evaluated with three final seeds on the same 640 official queries used for the dense and fusion comparisons.

For logits $z_{ij} = s(q_i, d_j)/\tau$, positive candidate positions P_i, candidate positions C_i, and

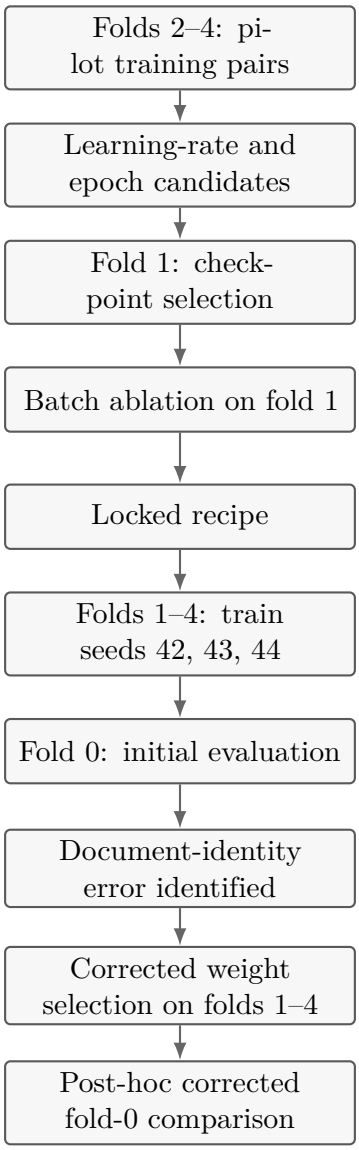


Figure 4.3: Training and evaluation evidence flow.

batch size B, the multi-positive loss is

$$\mathcal{L} = -\frac{1}{B}\sum_{i=1}^{B}\left[\log \sum_{j\in P_i} \exp(z_{ij}) - \log \sum_{j\in C_i} \exp(z_{ij})\right].$$

Study B is reported separately from Study A because it uses a different recipe and answers a different question: whether dense-only gains transfer into complete hybrids.

4.8 Method Summary

The methodology fixes the retrieval task at article ranking, preserves query and corpus identities, compares lexical and dense signals under selected fusion weights, and evaluates legal-domain adaptation only after explicit selection boundaries. BCA adds a dataset-construction and evaluation-design contribution by exposing coverage and component-dependence limits. Chapter 5 reports the resulting effectiveness, failure analysis, and adaptation outcomes.

Chapter 5

Experimental Evaluation

This chapter reports the experiments in the order of the research questions. It first evaluates lexical and hybrid retrieval, then isolates the incremental contribution of E5, then examines BCA coverage and component-aware reliability, and finally analyzes legal-domain adaptation. The central adaptation finding is that dense retriever improvement does not necessarily transfer into a complete hybrid retrieval system.

5.1 Datasets and Experimental Setup

Quantitative table values are displayed to four decimal places, using scientific notation with four fractional digits in the mantissa for tiny nonzero values. Deltas are calculated before display rounding; integer counts and indices remain integers.

The evaluated conditions and corpus sizes are defined in Table 4.1. ALQAC and Zalo each have clean and parsed corpus representations; their A/B conditions share query identities and are therefore paired corpus conditions rather than independent datasets. BCA is evaluated separately because many questions share legal targets and because some targets are unavailable in the indexed corpus.

NDCG@10 is the primary endpoint. Hit@10, MRR@10, and MAP@10 are reported to distinguish top-ten recovery, first-relevant ordering, and multiple-target ranking. ALQAC and Zalo use query-level paired comparisons for the two-way versus three-way hybrid. BCA uses component-aware analysis in addition to descriptive query-level summaries. These units of inference are not interchangeable.

Table 5.1: System identities and comparison boundaries.

Configuration	**Baseline role**	**Adaptation role**
BM25	Fixed lexical reference	Unchanged lexical component
BGE-M3 dense	Pretrained dense retriever	Adapted in the legal-domain fine-tuning studies
E5 dense	Additional pretrained dense retriever	Frozen throughout the thesis

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Configuration	Baseline role	Adaptation role
BM25+BGE-M3	Two-way lexical–semantic hybrid	Evaluated with pretrained and adapted BGE-M3
BM25+ BGE-M3+E5	Three-way hybrid	Evaluated with pretrained and adapted BGE-M3 while E5 remains fixed

The experiments compare methods within their own query populations and selection boundaries. Results from different folds, corpora, or adaptation recipes are not pooled into a single cross-protocol score.

5.2 Lexical and Hybrid Retrieval

This section compares lexical–semantic hybrid retrieval with the fixed BM25 reference. Table 5.2 reports BM25, pretrained dense retrievers, and hybrid systems for ALQAC and Zalo. Table 5.3 summarizes the hybrid changes relative to BM25 and includes BCA's combined descriptive view.

Table 5.2: Pretrained standalone and hybrid results, query-macro OOF population.

Condition	Method	NDCG@10	MRR@10	MAP@10	Hit@10
ALQAC A	BM25	0.7986	0.7648	0.7621	0.9112
ALQAC A	BGE-M3 pretrained	0.8437	0.8134	0.8108	0.9433
ALQAC A	E5 pretrained	0.8126	0.7729	0.7722	0.9371
ALQAC A	BM25+BGE-M3	0.8805	0.8534	0.8516	0.9692
ALQAC A	BM25+BGE-M3+ E5	0.8784	0.8519	0.8504	0.9630
ALQAC B	BM25	0.6479	0.6175	0.6159	0.7472
ALQAC B	BGE-M3 pretrained	0.6684	0.6401	0.6382	0.7620
ALQAC B	E5 pretrained	0.6510	0.6177	0.6165	0.7583
ALQAC B	BM25+BGE-M3	0.7067	0.6812	0.6797	0.7904
ALQAC B	BM25+BGE-M3+ E5	0.7055	0.6805	0.6793	0.7867
Zalo A	BM25	0.5264	0.4731	0.4682	0.7100
Zalo A	BGE-M3 pretrained	0.6424	0.5917	0.5850	0.8235

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Condition	Method	NDCG@10	MRR@10	MAP@10	Hit@10
Zalo A	E5 pretrained	0.5640	0.5104	0.5046	0.7509
Zalo A	BM25+BGE-M3	0.6805	0.6298	0.6235	0.8583
Zalo A	BM25+BGE-M3+ E5	0.6818	0.6317	0.6254	0.8576
Zalo B	BM25	0.5620	0.5099	0.5050	0.7406
Zalo B	BGE-M3 pretrained	0.6889	0.6431	0.6366	0.8526
Zalo B	E5 pretrained	0.6259	0.5748	0.5689	0.8054
Zalo B	BM25+BGE-M3	0.7214	0.6767	0.6705	0.8789
Zalo B	BM25+BGE-M3+ E5	0.7344	0.6913	0.6845	0.8889

BGE-M3 is the strongest individual retriever by NDCG@10 in all four ALQAC/Zalo conditions. Both hybrids exceed BM25, and both also exceed the strongest individual component in these conditions. The result supports lexical–semantic complementarity at the system level, although it does not identify the linguistic mechanism behind each gain.

Table 5.3: Hybrid NDCG@10 changes relative to fixed BM25.

Condition	Hybrid	BM25	Hybrid score	Absolute change	Relative change
ALQAC A	BM25+ BGE-M3	0.7986	0.8805	+0.0819	+10.2590%
ALQAC A	BM25+ BGE-M3+E5	0.7986	0.8784	+0.0798	+9.9970%
ALQAC B	BM25+ BGE-M3	0.6479	0.7067	+0.0588	+9.0690%
ALQAC B	BM25+ BGE-M3+E5	0.6479	0.7055	+0.0576	+8.8910%
Zalo A	BM25+ BGE-M3	0.5264	0.6805	+0.1541	+29.2760%
Zalo A	BM25+ BGE-M3+E5	0.5264	0.6818	+0.1554	+29.5210%
Zalo B	BM25+ BGE-M3	0.5620	0.7214	+0.1594	+28.3680%
Zalo B	BM25+ BGE-M3+E5	0.5620	0.7344	+0.1724	+30.6830%
BCA combined descriptive	BM25+ BGE-M3	0.1687	0.2283	+0.0597	+35.3820%

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Condition	Hybrid	BM25	Hybrid score	Absolute change	Relative change
BCA combined descriptive	BM25+ BGE-M3+E5	0.1687	0.2297	+0.0610	+36.1760%

The hybrid gains over BM25 are large in Zalo and smaller but still positive in ALQAC. BCA also shows a positive descriptive difference from BM25, but its lower absolute scores must be interpreted with coverage limits and component dependence. The answer to RQ1 is therefore positive within the evaluated conditions: lexical–semantic hybrids improve the fixed BM25 reference. The result should not be generalized into a claim that one hybrid configuration will transfer unchanged to every legal corpus.

5.3 Incremental Contribution of E5

This section tests whether E5 adds value beyond BM25+BGE-M3. This is a stricter question than asking whether E5 is a competent standalone retriever. The relevant comparison is the two-way hybrid against the three-way hybrid.

Table 5.4: Incremental E5 contribution on the canonical ALQAC and Zalo populations.

Condition	Two-way NDCG	Three-way NDCG	Absolute change	Relative change	Stored Wilcoxon p
ALQAC A	0.8805	0.8784	-0.0021	-0.2380%	0.4498
ALQAC B	0.7067	0.7055	-0.0012	-0.1630%	0.7449
Zalo A	0.6805	0.6818	+0.0013	+0.1900%	0.1548
Zalo B	0.7214	0.7344	+0.0130	+1.8030%	5.1598×10^{-7}

The incremental E5 result is condition-dependent. ALQAC shows small negative NDCG@10 differences in both corpus representations. Zalo A shows a very small positive difference, while Zalo B shows the clearest positive result: an absolute NDCG@10 gain of approximately 0.0130, with higher MRR@10 and Hit@10 as well. The p-values in Table 5.4 are reported as stored query-level paired tests; they do not remove the need to interpret multiple conditions and possible query dependence carefully.

The A/B corpus comparison also resists a simple parser narrative. ALQAC has substantially lower hybrid scores in the parsed condition than in the clean condition. Zalo, in contrast, has higher scores in the parsed-with-fallback condition than in the clean condition. This does not imply that parsing is inherently harmful for ALQAC and inherently beneficial for Zalo. The two collections differ in construction, target mapping, fallback behavior, and

article universe. A matched no-fallback ablation was not available, so the independent effect of fallback remains unmeasured.

The answer to RQ2 is therefore not a general yes. E5 provides useful incremental value in the parsed Zalo condition, negligible or unsupported value in clean Zalo, and no observed improvement in ALQAC. This is one of the thesis's main practical findings: adding a dense retriever expands the search space, but it does not guarantee better ranking on unseen data.

5.4 BCA Coverage and Component-Aware Evaluation

BCA serves two purposes. It extends the evaluation beyond ALQAC and Zalo, and it exposes how corpus coverage and target dependence affect interpretation. Table 5.5 reports BCA results under the giant-component, small-component, and combined descriptive views.

Table 5.5: Compact BCA protocol and coverage views. All metrics are at cutoff 10; OOF denotes out-of-fold.

Protocol	Cohort	Method	N	NDCG	MRR	MAP	Hit
Giant	All	BM25	484	0.1670	0.2469	0.1098	0.4545
Giant	All	BGE-M3	484	0.2153	0.3143	0.1453	0.5455
Giant	All	BM25+BGE	484	0.2275	0.3414	0.1573	0.5517
Giant	All	BM25+BGE+E5	484	0.2295	0.3413	0.1560	0.5702
Giant	Answerable	BM25	394	0.2051	0.3033	0.1348	0.5584
Giant	Answerable	BGE-M3	394	0.2645	0.3861	0.1785	0.6701
Giant	Answerable	BM25+BGE	394	0.2795	0.4194	0.1932	0.6777
Giant	Answerable	BM25+BGE+E5	394	0.2819	0.4193	0.1916	0.7005
Small OOF	All	BM25	62	0.1817	0.2157	0.1382	0.3387
Small OOF	All	BGE-M3	62	0.2015	0.2392	0.1597	0.3387
Small OOF	All	BM25+BGE	62	0.2346	0.2921	0.1865	0.3871
Small OOF	All	BM25+BGE+E5	62	0.2309	0.2838	0.1815	0.3871
Small OOF	Answerable	BM25	46	0.2449	0.2907	0.1863	0.4565
Small OOF	Answerable	BGE-M3	46	0.2715	0.3225	0.2153	0.4565
Small OOF	Answerable	BM25+BGE	46	0.3162	0.3937	0.2514	0.5217
Small OOF	Answerable	BM25+BGE+E5	46	0.3112	0.3825	0.2446	0.5217
Combined OOF	All	BM25	546	0.1687	0.2434	0.1130	0.4414
Combined OOF	All	BGE-M3	546	0.2137	0.3058	0.1469	0.5220
Combined OOF	All	BM25+BGE	546	0.2283	0.3358	0.1606	0.5330
Combined OOF	All	BM25+BGE+E5	546	0.2297	0.3348	0.1589	0.5495
Combined OOF	Answerable	BM25	440	0.2093	0.3020	0.1402	0.5477
Combined OOF	Answerable	BGE-M3	440	0.2652	0.3795	0.1823	0.6477
Combined OOF	Answerable	BM25+BGE	440	0.2833	0.4167	0.1993	0.6614
Combined OOF	Answerable	BM25+BGE+E5	440	0.2850	0.4154	0.1972	0.6818

The BCA reference graph contains one giant component with 484 of 546 queries and 49 smaller components containing the remaining 62 queries. This structure affects statistical

interpretation. The combined descriptive view shows the three-way hybrid slightly above the two-way hybrid in NDCG@10, but the small-component out-of-fold view shows a negative NDCG@10 difference for the three-way system. The corresponding component-aware inference does not establish an E5 advantage.

Coverage is also a major limitation. Of the 546 BCA queries, 440 have at least one represented target and 106 have none. A further 229 answerable queries are partial-target cases in which some, but not all, targets are present. An unanswerable query cannot receive a positive Hit@10 from any ranking method over the fixed corpus. Therefore, BCA's lower end-to-end scores partly reflect corpus and mapping limitations rather than only scoring failure.

The BCA result is a dataset and evaluation contribution rather than a simple model leaderboard. It shows that an end-to-end legal retrieval score can mix three phenomena: whether the evidence exists, whether it is mapped to the correct legal identity, and whether the retriever ranks it highly. A method improvement can address only the third problem unless corpus construction changes as well.

5.5 Failure Analysis

The failure analysis asks what a model intervention could plausibly repair. Missing targets require corpus or mapping work. Available targets outside the top ten require ranking or fusion improvements. Relevant articles demoted within the top ten affect NDCG and MRR even when Hit@10 is unchanged. These distinctions guide the adaptation experiment but do not prove that fine-tuning will solve the observed failures.

BCA provides the clearest evidence of non-rankable failures: 106 questions have no mapped target in the corpus. For these questions, changing BM25 weights, dense encoders, or fusion rules cannot recover the target. Partial-target queries create a second limitation: a system may retrieve an available relevant article while still being penalized for missing other targets that are absent from the corpus.

For ALQAC and Zalo, top-ten boundary cases show that E5 can both recover and lose hits. The same added signal that improves parsed Zalo on average can demote specific relevant articles below the cutoff. Appendix E preserves representative cases. The main conclusion is more general: aggregate improvement does not imply uniform query-level improvement, and aggregate regression does not imply that every query becomes worse.

The analysis remains an information-retrieval analysis. It identifies target absence, top-ten transitions, within-law article demotion, and persistent misses. It does not assign legal-reasoning causes such as misunderstanding exceptions or temporal validity. A legally adjudicated error taxonomy would require a sampling plan, full text review, annotation rules, and reviewer agreement. That study was not performed here.

5.6 Legal-Domain Adaptation

This section examines whether legal-domain BGE-M3 adaptation improves retrieval beyond development selection. The narrative has six parts: Study A development selection, Study A corrected hybrid evaluation on unseen data, Study B dense-only adaptation, transfer of dense improvements into hybrid retrieval, query-level analysis, and overall interpretation.

5.6.1 Development Selection

Study A: Hybrid-Oriented Adaptation selects a BGE-M3 training recipe on development queries. Table 5.6 reports the learning-rate and epoch pilot under passage length 512. Table 5.7 reports the subsequent batch ablation at the selected learning rate and epoch.

Table 5.6: Passage-512 learning-rate and epoch pilot: development NDCG@10.

Learning rate	**Epoch**	**Two-way hybrid**	**Three-way hybrid**
5.0000×10^{-6}	1	0.7470	0.7685
5.0000×10^{-6}	2	0.7603	0.7785
5.0000×10^{-6}	3	0.7647	0.7815
1.0000×10^{-5}	1	0.7509	0.7816
1.0000×10^{-5}	2	0.7658	0.7873
1.0000×10^{-5}	3	0.7628	0.7897
2.0000×10^{-5}	1	0.7440	0.7757
2.0000×10^{-5}	2	0.7691	0.7923
2.0000×10^{-5}	3	0.7689	0.7958

The pilot shows development improvements across several candidate settings. The selected recipe uses learning rate 0.00002 and two epochs. The choice is not evidence that the selected point is a global optimum; it is the best supported point under the restricted search and selection rule.

Table 5.7: Batch ablation at the selected learning rate and epoch; scores are development NDCG@10.

Batch/accum.	**Two-way**	**Δ**	**Three-way**	**Δ**
Pretrained baseline	0.7172	–	0.7245	–
1 / 32	0.7725	+0.0553	0.7869	+0.0624
2 / 16	0.7693	+0.0521	0.7924	+0.0679
4 / 8	0.7781	+0.0609	0.7934	+0.0689

The selected development configuration improves two-way NDCG@10 by 0.060945 and three-way NDCG@10 by 0.068882. This justifies testing the intervention on unseen data. It does not establish generalization, because the development queries are part of model selection.

5.6.2 Corrected Hybrid Evaluation on Unseen Data

The corrected comparison on unseen data evaluates Zalo parsed fold 0 with 640 queries. Fusion weights are fitted on the permitted training-side folds using pretrained BGE-M3 and then held fixed when BGE-M3 is replaced by its adapted version. This design measures encoder replacement within a specified hybrid rather than unrestricted retuning of the whole system.

The corrected two-way weights are BM25/BGE/E5 = 0.2/0.8/0.0. The corrected three-way weights are 0.2/0.5/0.3. Because a flawed preliminary analysis had already exposed the same fold, the corrected comparison is interpreted as a post-hoc evaluation on data unseen during training and selection, not as a pristine first-use test.

Table 5.8: Corrected Zalo fold-0 two-way hybrid evaluation; all metrics are at cutoff 10.

BGE state	**Seed**	**NDCG**	**Δ**	**MRR**	**MAP**	**Hit**
Pretrained	Fixed	0.7516	–	0.7015	0.6985	0.9141
Fine-tuned	42	0.7005	−0.0511	0.6470	0.6451	0.8750
Fine-tuned	43	0.7005	−0.0511	0.6490	0.6467	0.8688
Fine-tuned	44	0.6977	−0.0539	0.6398	0.6374	0.8859
Fine-tuned	Mean	0.6995	−0.0520	0.6453	0.6430	0.8766

All three two-way seeds score below the pretrained two-way baseline. The mean NDCG@10 change is -0.052045, and MRR@10, MAP@10, and Hit@10 are also lower. The result therefore cannot be rescued by choosing a different seed from the three reported runs.

Table 5.9: Corrected Zalo fold-0 three-way hybrid evaluation; all metrics are at cutoff 10.

BGE state	**Seed**	**NDCG**	**Δ**	**MRR**	**MAP**	**Hit**
Pretrained	Fixed	0.7595	–	0.7077	0.7048	0.9266
Fine-tuned	42	0.7320	−0.0275	0.6771	0.6743	0.9141
Fine-tuned	43	0.7340	−0.0255	0.6771	0.6747	0.9203
Fine-tuned	44	0.7378	−0.0217	0.6794	0.6771	0.9281
Fine-tuned	Mean	0.7346	−0.0249	0.6779	0.6754	0.9208

The three-way hybrid also has lower mean NDCG@10 after fine-tuning. The regression is smaller than in the two-way system, but it remains a regression against the matched pretrained three-way comparator. Seed 44 slightly improves Hit@10 while reducing

NDCG@10 and MRR@10, showing why the primary rank-sensitive metric must remain central.

Table 5.10: Corrected paired statistics for NDCG@10.

Comparison	Queries	Clusters	Mean Δ	95% cluster CI	Raw p	Holm p
Fine-tuned minus pretrained, two-way	640	106	−0.0520	[−0.0919, +0.1076]	0.7480	1.0000
Fine-tuned minus pretrained, three-way	640	106	−0.0249	[−0.0542, +0.0938]	0.7480	1.0000

The intervals in Table 5.10 include zero, so the observed regressions are not described as statistically established population-level harm. However, non-significance is not equivalence, and the practical improvement threshold of +0.01 NDCG@10 is not met. The correct conclusion is that the selected adaptation recipe failed to demonstrate hybrid improvement on unseen data under this protocol.

5.6.3 Study B: Dense-Oriented Adaptation

Study A does not by itself determine whether adaptation failed at the dense-retriever level or only after integration into the hybrid. Study B directly evaluates dense-only BGE-M3 adaptation before testing hybrid transfer.

Table 5.11: Study B post-hoc dense benchmark on 640 official queries.

System	NDCG@10	Delta NDCG@10	MRR@10	Recall@100
Pretrained	0.6740	0.0000	0.6213	0.9617
Fine-tuned seed 42	0.6932	0.0192	0.6448	0.9531
Fine-tuned seed 43	0.6925	0.0185	0.6431	0.9516
Fine-tuned seed 44	0.6928	0.0189	0.6437	0.9531
Fine-tuned seed mean	0.6928	0.0189	0.6438	0.9526

Study B shows a mean NDCG@10 gain of 0.018866, with all three seeds improving this endpoint. Mean Recall@100 decreases by 0.009115. The dense result therefore supports the claim that legal-domain fine-tuning can improve the dense encoder on this fixed-corpus query benchmark, while also showing a small recall tradeoff at depth 100.

5.6.4 Transfer of Dense Improvements into Hybrid Retrieval

The crucial question is whether the dense improvement remains useful inside the full hybrid. Table 5.12 reports the Study B primary locked-weight hybrid outcomes. Fusion

weights are selected with pretrained BGE-M3 and then held fixed for the corresponding fine-tuned comparison.

Table 5.12: Study B primary fusion outcomes under pretrained-locked weights.

Hybrid	Pretrained	FT mean	Δ NDCG	Rel. Δ	Δ R@100	Positive seeds	Gate
Two-way	0.7526	0.7483	-0.0043	-0.5707%	-0.0057	0/3	Fail
Three-way	0.7467	0.7476	0.0009	0.1203%	-0.0031	2/3	Fail

The dense-only gain does not transfer into a practically useful hybrid gain. The two-way hybrid decreases by 0.004295 NDCG@10. The three-way hybrid increases by only 0.000898 NDCG@10, far below the +0.01 improvement threshold. Thus, Study B sharpens the main RQ3 finding: adaptation can improve standalone dense ranking while failing to improve the complete hybrid retrieval system by a meaningful amount.

The mechanism is not isolated by these experiments. Possible explanations include changed score calibration, overlap between lexical and dense evidence, and fusion weights selected for pretrained scores. Because these mechanisms are not independently varied, they remain hypotheses for future work rather than established causes.

5.6.5 Query-Level Analysis

Aggregate means can hide mixtures of repaired and introduced failures. Table 5.13 summarizes the Study A corrected seed-42 hybrid changes. The first three rows partition all 640 queries by NDCG@10 change; the final rows describe top-ten transitions and are not additional disjoint totals.

Table 5.13: Query-level changes after fine-tuning, corrected seed 42.

Outcome	Two-way hybrid	Three-way hybrid
Higher NDCG@10	102	102
Lower NDCG@10	210	163
Unchanged NDCG@10	328	375
Previously missing at ten, now retrieved	23	16
Previously retrieved at ten, now missing	48	24
Missing at ten in both systems	32	31

The two-way system repairs some top-ten misses, but introduces more than twice as many new top-ten misses as it repairs. The three-way system introduces fewer new misses, consistent with its smaller aggregate regression, but it still introduces more misses than it

repairs. These counts show why the negative mean should not be interpreted as uniform deterioration, and why isolated repaired examples cannot overturn the aggregate result.

Table 5.14: Study B NDCG changes by query, using mean fine-tuned seed metric minus pretrained.

Comparison	**Improved**	**Worse**	**Tied**
Dense	127	96	417
Two-way	81	93	466
Three-way	85	86	469

Study B's dense-only comparison has more improved than worsened queries, while the Study B hybrids do not show the same favorable balance. This query-level view is consistent with the aggregate finding: dense improvement exists, but much of it is absorbed or neutralized when the adapted encoder is placed inside the full fusion system.

5.6.6 Overall Interpretation

The RQ3 evidence has a clear sequence. First, Study A's development selection shows large gains and justifies testing the adaptation. Second, Study A's corrected hybrid evaluation on unseen data shows regressions for both two-way and three-way hybrids. Third, Study B shows that legal fine-tuning can improve dense BGE-M3 ranking. Fourth, that dense gain largely fails to transfer into locked-weight hybrid retrieval.

This sequence supports a stronger and more nuanced conclusion than either "fine-tuning works" or "fine-tuning fails." It works at the dense-retriever level in Study B, but it does not necessarily improve the complete lexical–semantic system. The complete system must therefore be evaluated directly. Development scores and dense-only scores are insufficient substitutes for end-to-end hybrid evaluation.

Several boundaries remain. Cross-collection adaptation to ALQAC and BCA was not established. Secondary retuned-weight hybrid results were not available, so the thesis does not claim that the adapted encoder could not be made useful under a different fusion policy. A fresh untouched cohort would be required for a cleaner confirmatory test. These limitations narrow the conclusion without weakening the observed negative transfer result.

5.7 Experimental Synthesis

The experiments answer the research questions as follows. For RQ1, hybrid retrieval improves the fixed BM25 reference in the evaluated ALQAC, Zalo, and descriptive BCA conditions. For RQ2, E5 has condition-dependent incremental value: parsed Zalo provides

the clearest positive case, while ALQAC and BCA do not support a general three-way advantage. For RQ3, legal-domain adaptation improves development scores and can improve dense-only ranking, but it does not necessarily improve the complete hybrid retrieval system.

The BCA analysis adds a methodological lesson. Evaluation cannot be reduced to a ranking table when target coverage and query dependence are substantial. Missing targets, partial target availability, and giant connected components affect what a retrieval score means. Reporting both end-to-end and answerable-only views makes this limitation visible rather than hiding it inside an average.

The principal scientific takeaway is that legal retrieval systems must be evaluated at the level at which they will be used. A model can be better in isolation while adding little to a hybrid; an added retriever can help one corpus representation and not another; a retrieval failure can be caused by absent evidence rather than poor ranking. These findings justify the thesis's emphasis on controlled comparison, negative results, and careful limitation of causal claims.

Chapter 6

Conclusion and Future Work

This chapter synthesizes the experimental findings rather than repeating the numerical analysis from Chapter 5. The conclusions distinguish improvement over BM25, incremental value beyond an existing hybrid, and transfer from dense adaptation into a complete retrieval system.

6.1 Summary of Findings

6.1.1 Hybrid Retrieval over the Lexical Baseline

The experiments support the value of lexical–semantic hybrid retrieval over the fixed BM25 reference in the evaluated conditions. Both the BM25+BGE-M3 hybrid and the BM25+BGE-M3+E5 hybrid improve NDCG@10 over BM25 across the reported ALQAC and Zalo conditions, and BCA shows the same direction in its combined descriptive view. The result is strongest as an empirical finding about these collections and corpus representations. It does not imply that a single weight setting, corpus construction policy, or hybrid architecture will transfer unchanged to other legal corpora.

6.1.2 Incremental Contribution of E5

The incremental value of E5 is condition-dependent. The clearest positive result occurs on parsed Zalo, where adding E5 increases NDCG@10 by approximately 0.0130. ALQAC shows small negative differences, clean Zalo shows only a small positive difference, and BCA does not establish a reliable small-component advantage. Therefore, the thesis does not support a universal claim that three-way retrieval is better than two-way retrieval. The value of the additional dense retriever must be evaluated against a strong two-way reference in the target corpus condition.

6.1.3 Effect of Legal-Domain Adaptation

The answer is mixed and is one of the main findings of the thesis. Study A: Hybrid-Oriented Adaptation shows substantial development-selection gains, but its corrected hybrid evaluation on data unseen during training and selection shows mean NDCG@10

decreases of 0.0520 for the two-way hybrid and 0.0249 for the three-way hybrid. Study B: Dense-Oriented Adaptation shows that fine-tuning can improve BGE-M3 as a dense retriever, increasing mean NDCG@10 by 0.0189. However, that improvement largely disappears after the adapted encoder is integrated into locked-weight hybrids. Thus, dense adaptation can improve an encoder without improving the complete lexical–semantic retrieval system by a meaningful amount.

6.2 Research Contributions

The first contribution is a controlled empirical evaluation of lexical–semantic hybrid retrieval for Vietnamese legal information retrieval. The thesis compares BM25, BGE-M3, E5, two-way fusion, and three-way fusion across multiple legal collections and corpus representations. It separates improvement over a lexical baseline from incremental improvement over an already strong hybrid.

The second contribution is an evaluation of legal-domain BGE-M3 adaptation at both dense-retriever and complete-hybrid levels. The work preserves negative and non-transfer results rather than treating them as failed experiments. The distinction between dense-only improvement and hybrid-system improvement is practically important for legal retrieval systems, where the deployed system is usually a pipeline rather than a single encoder.

The third contribution is a dataset and evaluation contribution involving the BCA collection. The thesis constructs a retrieval collection from public questions and published answer citations, analyzes target coverage, and applies component-aware evaluation. This contribution shows that corpus availability and query dependence can materially affect how legal retrieval metrics should be interpreted.

6.3 Discussion and Limitations

The experiments show that hybrid retrieval is useful, but they also show why retrieval evaluation should not stop at a favorable aggregate score. A hybrid can improve over BM25 while an additional component contributes little. A model can repair some queries while introducing other failures. A benchmark can contain many questions while having fewer independent legal components than the query count suggests. These distinctions matter in legal retrieval because a system must retrieve specific provisions, not merely topically related documents.

The E5 result illustrates the limits of model-count reasoning. Adding another dense retriever increases representational diversity and expands the fusion search space, but the measured benefit depends on the collection and corpus representation. Parsed Zalo

benefits most clearly; ALQAC does not. This finding supports local validation rather than a generic architectural rule.

The adaptation results illustrate the limits of development selection. Study A's fine-tuning recipe performs well on development queries but regresses in corrected hybrid evaluation on unseen data. Study B shows that this is not simply because legal-domain fine-tuning is incapable of improving dense retrieval. Instead, the complete hybrid can absorb, dilute, or counteract dense improvements because fusion weights, lexical evidence, and score calibration all interact. The present experiments do not isolate which mechanism dominates.

The publicly available Vietnamese legal data suitable for supervised retrieval adaptation remain limited in the present setting. Additional diverse and carefully validated query–article training pairs may improve the fine-tuning process, particularly by providing broader coverage of legal topics and relevance relations. However, the study did not evaluate this intervention; it is therefore a hypothesis for future work rather than an empirical conclusion.

BCA exposes dataset limitations that are not model limitations. Some targets are absent from the indexed corpus, and some answerable queries are only partially covered. A ranking model cannot retrieve evidence that is not present. Therefore, end-to-end scores should be interpreted as measuring the whole retrieval pipeline, including corpus construction and target mapping, not only scoring quality.

Several validity limits remain. The evaluation is restricted to Vietnamese legal collections and recorded relevance judgments. It does not establish multilingual generalization, cross-jurisdiction transfer, downstream legal-answer correctness, current legal validity, user satisfaction, latency, throughput, or deployment cost. Study A's corrected adaptation evaluation on data unseen during training and selection is post-hoc because it follows a flawed preliminary analysis of that same evaluation fold. Study B also uses a fixed-corpus benchmark with prior test observation, so it should not be described as a pristine prospective confirmation.

Statistical conclusions are also bounded. Non-significant intervals do not prove equivalence, and detectable differences do not automatically imply practical importance. The thesis therefore distinguishes observed mean changes, practical improvement thresholds, and statistical uncertainty.

6.4 Future Work

A new, prospectively controlled evaluation should be conducted on an untouched query set. Corpus construction, model selection, fusion configuration, and statistical procedures should be fixed before examining the final test results.

Corpus quality should also be improved, particularly for the BCA collection. Coverage gaps can be addressed through version-aware mapping of legal documents, while query-level failures can be systematically sampled and annotated using a defined rubric. This would help distinguish missing evidence, annotation limitations, lexical mismatch, semantic confusion, and fusion-related ranking errors.

Future work should further extend the evaluation from retrieval performance to the complete RAG system. This includes assessing answer correctness, citation support, legal sufficiency, latency, throughput, and computational cost. Retrieval metrics remain useful indicators, but they cannot replace direct evaluation of generated legal answers and deployed system performance.

6.5 Closing Remarks

This thesis shows that Vietnamese legal retrieval benefits from lexical–semantic hybridization, but also that more components or better dense-only scores do not automatically produce a better complete system. The most important lesson is methodological: legal retrieval claims must be tied to the exact corpus, target coverage, selection boundary, and system level being evaluated. A disciplined negative result can be as informative as a positive result when it prevents an unsupported deployment claim.

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